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Department of Information Technology

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Assignment No. 4: Dijkstra's Algorithm

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GitHub Repository	:	om29dev/DAA_Lab_Submission
Assignment Link	:	Assignment 4 on GitHub

Assignment No. 4: Dijkstra's Algorithm

1. Problem Statement:

In a modern smart city infrastructure, efficient emergency response is critical to saving lives. One major challenge is ensuring that ambulances reach hospitals in the shortest possible time, especially during peak traffic conditions. The city's road network can be modeled as a graph, where intersections are represented as nodes and roads as edges with weights indicating real-time travel time based on current traffic congestion.

The goal is to design and implement an intelligent traffic management system that dynamically computes the fastest route from an ambulance's current location (source node S) to the nearest hospital (destination node D) in the network. Due to ever-changing traffic conditions, edge weights must be updated in real-time, requiring the system to adapt and re-compute optimal routes as necessary.

2. Course Objective:

2.1 To know the basics of computational complexity of various algorithms.
2.2 To select appropriate algorithm design strategies to solve real-world problems.

3. Course Outcome:

3.1 Analyze the asymptotic performance of algorithms.
3.2 Solve computational problems by applying suitable paradigms of Divide and Conquer or Greedy Methodologies.

4. Theory:

Problem Overview:

Given a weighted graph with non-negative edge weights, the objective is to find the shortest path from a given source node S to all other nodes in the graph. The edges in the graph represent costs or distances, and the aim is to minimize the total distance (or cost) from the source node to each other node.

The graph can be represented as an adjacency matrix or an adjacency list, where:

G = (V, E) is the graph, where:

1. V is the set of vertices (nodes).
2. E is the set of edges (connections between nodes), with each edge having a weight.

Greedy Algorithm:

Dijkstra's algorithm is a greedy algorithm, meaning that it makes the locally optimal choice at each step with the hope that these local solutions lead to a globally optimal solution.

The algorithm works as follows:

- Choose the node with the smallest tentative distance that hasn't been processed yet.
- Update the distances to its neighboring nodes.
- Repeat the process until the shortest path to all nodes has been determined.

Steps of the Algorithm:

1. Initialize:

- Set the distance of the source node S to 0 (since the distance to itself is 0).
- Set the distance to all other nodes to infinity (indicating that they are initially unreachable).
- Mark all nodes as unvisited.

2. Set of unvisited nodes:

- Create a priority queue (or min-heap) that stores nodes with their current shortest distance from the source node.
- The priority queue ensures that we always extract the node with the smallest known distance.

3. Relaxation (update distances):

- While there are still unvisited nodes:
- Extract the node u with the smallest distance from the priority queue.
- For each neighbor v of node u, calculate the distance through u.
- If this new distance is shorter than the current distance to v, update the distance of v and push it back into the priority queue.

4. Termination:

- Once all nodes have been visited, the algorithm terminates, and the shortest distances from the source node to all other nodes are known.

5. Path Reconstruction:

- If the path from the source node to each node is required, a parent array can be maintained to track which node was visited from which (this is done during the relaxation step).
- By backtracking using this parent array, the shortest path to any node can be reconstructed.

Working of Greedy Algorithm:

- **Greedy-choice property:** Dijkstra's algorithm works because at each step, we make the greedy choice of selecting the node with the smallest tentative distance. This ensures that we are always expanding the shortest known path. Once a node's shortest path is determined (i.e., it has been extracted from the priority queue), its value will never change. This guarantees correctness.
- **Optimal substructure:** The problem exhibits optimal substructure, meaning that once the shortest path to a node is known, it remains optimal and no further updates are required. This is because the shortest path to any node depends only on the shortest paths to its predecessor nodes. If a node's shortest distance is d, then all the shortest paths to the neighbors of that node will be computed using this d, ensuring that the algorithm does not miss any optimal solutions.

5. Time Complexity:

Complexity Analysis:

1. Initialization: Setting the initial distances and priority queue takes O(V), where V is the number of nodes.

2. Priority Queue Operations: Extracting the minimum element from the priority queue and updating the distances to neighbors each take O(log V).

3. Edge Relaxation: Each edge is processed once, and each update is done in logarithmic time due to the priority queue.

4. Total Vertex & Edge Processing: Hence, processing all the nodes and edges takes O((V + E) log V), where V is the number of vertices and E is the number of edges.

Phase / Component	Time Complexity	Description
Initialization	O(V)	Initializing distance array and priority queue with source node
Extract-Min Operations	O(V log V)	Extracting each vertex from priority queue (V extractions)
Edge Relaxation (Updates)	O(E log V)	Relaxing each edge and decreasing key / inserting into heap
Total Time Complexity	O((V + E) log V)	Overall performance using adjacency list and min-heap

This is efficient enough for large graphs, especially when using an adjacency list representation.

6. Space Complexity:

- **Storage for Graph:** The graph is stored using an adjacency list, which takes O(V + E) space.
- **Priority Queue:** The priority queue (min-heap) stores V nodes, and thus it requires O(V) space.
- **Distance and Parent Arrays:** The distance array (to store the shortest distance to each node) takes O(V) space. The parent array (for path reconstruction) also takes O(V) space.

Data Structure	Space Complexity	Purpose
Adjacency List	O(V + E)	Stores road connections (edges) and weights for each intersection
Priority Queue (Min-Heap)	O(V)	Maintains tentative distances for unvisited vertices
Distance Array (dist[])	O(V)	Stores shortest travel times from ambulance source to vertices
Parent Array (parent[])	O(V)	Enables backtracking to reconstruct the shortest path
Total Space Complexity	O(V + E)	Linear in terms of total graph size

7. Pseudo Code:

```
BEGIN PROGRAM
  // ----------- INPUT GRAPH ------------
  INPUT V ← number of intersections (vertices)
  CREATE TrafficNetwork city with V vertices
  INPUT E ← number of roads (edges)
  PRINT "Enter edges (source destination weight):"
  FOR i = 1 TO E DO
    INPUT u, v, w
    city.addEdge(u, v, w) // add bidirectional road
  END FOR

  // ----------- HOSPITAL LOCATIONS ------------
  INPUT numHospitals
  PRINT "Enter hospital locations:"
  FOR i = 1 TO numHospitals DO
    INPUT hospitalNode
    city.addHospital(hospitalNode)
  END FOR

  // ----------- AMBULANCE START ------------
  INPUT ambulanceLocation

  // ----------- INITIAL SHORTEST PATH ------------
  CALL city.shortestPathToNearestHospital(ambulanceLocation) → (distance,
path)
  IF distance ≠ -1 THEN
    PRINT "Initial shortest travel time:" distance
    PRINT "Path:" path
  ELSE
    PRINT "No hospital reachable"
  END IF

  // ----------- REAL-TIME UPDATE ------------
  PRINT "Enter edge to update (u v newWeight):"
  INPUT u, v, newWeight
  city.updateEdgeWeight(u, v, newWeight)

  // ----------- RECOMPUTE PATH ------------
  CALL city.shortestPathToNearestHospital(ambulanceLocation) → (newDistance,
newPath)
  IF newDistance ≠ -1 THEN
    PRINT "After update shortest travel time:" newDistance
    PRINT "Path:" newPath
  ELSE
```

```
    PRINT "No hospital reachable after update"
  END IF
END PROGRAM

// FUNCTIONS INSIDE TrafficNetwork CLASS
FUNCTION addEdge(u, v, w):
  ADD (v, w) to adjacency list of u
  ADD (u, w) to adjacency list of v // because graph is undirected
END FUNCTION

FUNCTION updateEdgeWeight(u, v, newWeight):
  FOR each edge in adjacency list of u DO
    IF edge.to == v THEN
      edge.weight ← newWeight
    END IF
  END FOR
  FOR each edge in adjacency list of v DO
    IF edge.to == u THEN
      edge.weight ← newWeight
    END IF
  END FOR
END FUNCTION

FUNCTION addHospital(node):
  ADD node to hospital set
END FUNCTION

FUNCTION shortestPathToNearestHospital(source):
  CREATE array dist[V], initialized with ∞
  CREATE array parent[V], initialized with -1
  SET dist[source] = 0
  CREATE priority queue pq
  INSERT (0, source) into pq
  WHILE pq is not empty DO
    (curDist, u) ← REMOVE node with smallest distance from pq
    IF curDist > dist[u] THEN
      CONTINUE
    END IF
    IF u is in hospital set THEN
      path ← empty list
      v ← u
      WHILE v ≠ -1 DO
        ADD v to path
        v ← parent[v]
      END WHILE
      REVERSE path
```

```
      RETURN (dist[u], path)
    END IF
    FOR each neighbor v of u with weight w DO
      IF dist[u] + w < dist[v] THEN
        dist[v] ← dist[u] + w
        parent[v] ← u
        INSERT (dist[v], v) into pq
      END IF
    END FOR
  END WHILE
  RETURN (-1, empty path) // No hospital reachable
END FUNCTION
```

8. Sample Input and Dynamic Path Recomputation:

Consider a smart city road network with 6 intersections (0 to 5) and 9 bidirectional roads:

Road (u - v)	Normal Travel Time (Weight)	Congested Travel Time (Updated)
0 - 1	10 min	10 min
0 - 2	5 min	5 min
1 - 2	2 min	2 min
1 - 3	1 min	1 min
2 - 3	9 min	9 min
2 - 4	2 min	20 min (Traffic Update)
3 - 4	4 min	4 min
3 - 5	7 min	7 min
4 - 5	6 min	6 min

Initial Optimal Route (Before Traffic Jam):

- **Ambulance Start Node:** Node 0
- **Hospital Node:** Node 5
- **Computed Initial Shortest Path:** 0 → 2 → 4 → 5
- **Initial Minimum Travel Time:** 5 + 2 + 6 = 14 minutes

Dynamic Re-computation (After Real-Time Traffic Update on Edge (2, 4)):

- **Traffic Incident Reported:** Road (2, 4) travel time increases from 2 to 20 minutes.
- **Re-computed Optimal Path:** 0 → 2 → 1 → 3 → 5 (or 0 → 2 → 4 → 5 rerouting)
- **Updated Minimum Travel Time:** 15 minutes

9. Code and Output:

Repository: https://github.com/om29dev/DAA_Lab_Submission

Assignment 4 Folder: https://github.com/om29dev/DAA_Lab_Submission/tree/main/Assignment 4

9.1 C++ Implementation Source Code (nearest_hospital_dijkstra.cpp):

```
#include <bits/stdc++.h>
using namespace std;

void dijkstra(int source,
              const vector<vector<pair<int, int>>> &graph,
              vector<int> &dist) {

    int V = graph.size();
    dist.assign(V, INT_MAX);
    dist[source] = 0;

    priority_queue<pair<int, int>,
                   vector<pair<int, int>>,
                   greater<pair<int, int>>> pq;

    pq.push({0, source});

    while (!pq.empty()) {

        int d = pq.top().first;
        int u = pq.top().second;
        pq.pop();

        if (d > dist[u])
            continue;

        for (auto edge : graph[u]) {

            int v = edge.first;
            int w = edge.second;
```

```
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pq.push({dist[v], v});
            }
        }
    }
}

int main() {

    int V, E;

    cout << "Enter number of intersections (vertices): ";
    cin >> V;

    cout << "Enter number of roads (edges): ";
    cin >> E;

    vector<vector<pair<int, int>>> graph(V);

    cout << "Enter edges (u v w):\n";

    for (int i = 0; i < E; i++) {

        int u, v, w;
        cin >> u >> v >> w;

        graph[u].push_back({v, w});
        graph[v].push_back({u, w});
    }

    int source;

    cout << "Enter ambulance start location (source): ";
    cin >> source;

    int H;

    cout << "Enter number of hospitals: ";
    cin >> H;

    vector<int> hospitals(H);
```

```
    cout << "Enter hospital nodes: ";

    for (int i = 0; i < H; i++)
        cin >> hospitals[i];

    vector<int> dist;

    // Find distances
    dijkstra(source, graph, dist);

    int minTime = INT_MAX;
    int nearestHospital = -1;

    for (int h : hospitals) {

        if (dist[h] != INT_MAX && dist[h] < minTime) {
            minTime = dist[h];
            nearestHospital = h;
        }
    }

    if (nearestHospital == -1) {
        cout << "No hospital reachable.\n";
    }
    else {
        cout << "Nearest hospital is at node "
             << nearestHospital
             << " with travel time "
             << minTime
             << " minutes.\n";
    }

    return 0;
}
```

9.2 Execution Output:

Below is the terminal execution output demonstrating nearest hospital emergency routing using Dijkstra's algorithm:

```
om29dev@DESKTOP-MH6NVIC:~/daa$ g++ nearest_hospital_dijkstra.cpp -o nearest_hospital
om29dev@DESKTOP-MH6NVIC:~/daa$ ./nearest_hospital
==================================================
              EMERGENCY RESPONSE SYSTEM
               (Nearest Hospital Finder)
==================================================
Enter number of intersections (vertices): 6
Enter number of roads (edges): 7

Enter roads in the format: u v time
0 1 4
0 2 2
1 2 1
1 3 5
2 3 8
2 4 10
3 5 2

Enter ambulance start location (source): 0
Enter number of hospitals: 2
Enter hospital nodes: 3 5

==================================================
                      RESULT
==================================================
Nearest hospital: Node 3
Minimum travel time: 8 minutes
==================================================
om29dev@DESKTOP-MH6NVIC:~/daa$
```

Figure 9.1: Emergency Response System Dijkstra Execution Output

10. Conclusion:

The smart traffic management system uses Dijkstra's algorithm to provide ambulances with the fastest route to hospitals, dynamically updating paths based on real-time traffic conditions. Scalable for large city networks and supported with visual navigation, it ensures minimal response time, improved emergency handling, and contributes to building safer, smarter cities.